Ferromagnetic scar tower under bulk dephasing: notes on the results so far

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20 October 2026

Abstract

We study how the ferromagnetic quantum many-body scar tower of a spin-1/2 chain decays under uniform σ^z dephasing of rate γ, comparing the scar state $|-\rangle^{\otimes L}$ with two non-scar product states, the x-Néel and z-Néel states. With exact density-matrix evolution up to $L = 14$ and quantum trajectories up to $L = 20$, the $1/e$ rate of the scar's transverse magnetization becomes L-independent for $\gamma \gtrsim 0.05$, and its thermodynamic value Γ_∞ vanishes as $\gamma \to 0$, roughly as $\gamma^{1/2}$ for $0.02 \leq \gamma \leq 0.2$. The thermal x-Néel signal instead decays at the closed-system thermalization rate ≈ 4.8 for every $\gamma \leq 0.1$. At $\gamma \leq 0.007$ and fixed $L \leq 20$, the scar rate sits at the finite-size cap $\Gamma_e \approx \gamma(L-1)$. A single-kick model, in which every σ^z kick damages the coherence as one flip evolved with H does, reproduces the measured rate at every L and $\gamma \leq 0.05$ to within $\approx 5\%$, without fit parameters. The thermodynamic lifetime at small γ is therefore fixed by how the damage of a single kick grows in time; $L \leq 24$ does not yet decide between ballistic ($\Gamma_\infty \propto \gamma^{1/2}$) and diffusive ($\propto \gamma^{2/3}$) growth.

Contents

1 Model **2**

2 How the decay rates are computed **2**

3 Exact results **4**

4 Scar against thermal decay up to $L = 20$ **4**
4.1 States and signals 4
4.2 Scar 5
4.3 Thermal references 5

5 Small γ: the finite-size cap **7**

6 Single-kick test **10**

7 Summary and open points **10**

A Earlier results with the previous coefficients ($L \leq 10$) **11**
A.1 Scar oscillations and their decay 11
A.2 Scar versus non-scar probes 11
A.3 Liouvillian spectrum 12
A.4 Dependence on γ at fixed $L = 6$ 12

1 Model

Hamiltonian. Spin-1/2 ring of L sites (periodic boundaries), $S^\alpha_j = \sigma^\alpha_j/2$, bond singlet projector $R_{j,j+1} = \frac{1}{4} - \mathbf{S}_j \cdot \mathbf{S}_{j+1}$:

$$H = H_I + hS^z_{\rm tot}, \qquad H_I = \sum_j \Big[\sum_\alpha \big(a_\alpha R_{j,j+1}\sigma^\alpha_{j+2} + b_\alpha \sigma^\alpha_{j-1} R_{j,j+1}\big) + \sum_{\alpha\beta} c^{(j)}_{\alpha\beta}\, \sigma^\beta_{j-1} R_{j,j+1} \sigma^\alpha_{j+2} \Big]. \quad (1)$$

Every term of H_I annihilates the ferromagnetic multiplet $|\Phi_n\rangle \propto (S^-_{\rm tot})^n |\uparrow\cdots\uparrow\rangle$, $n = 0, \ldots, L$, so $H|\Phi_n\rangle = h(L/2 - n)|\Phi_n\rangle$: an equally spaced tower of $L+1$ exact scars.

Coefficients. $h = 1$, $a = b = (1,1,1)$, and $c_{xx} = c_{yy} = c_{zz} = 1$ (all other $c_{\alpha\beta} = 0$), each of the three multiplied on every bond j by an independent factor uniform in $[0.5, 1.5]$. The random c removes translation and reflection symmetry; a and b break SU(2), U(1), spin inversion and complex conjugation. One realization per L (seed $1000 + L$), shared by every run. All results in Secs. 4–6 use these coefficients; the earlier results with different coefficients are collected in Appendix A.

Dephasing.

$$\dot\rho = \mathcal{L}\rho = -i[H, \rho] + \gamma \sum_j \big(\sigma^z_j \rho\, \sigma^z_j - \rho\big). \quad (2)$$

In the Pauli basis $\rho = 2^{-L} \sum_P r_P P$ the dissipator is diagonal, $\mathcal{D}(P) = -2\gamma\, n_{xy}(P)\, P$, with $n_{xy}(P)$ the number of sites carrying σ^x or σ^y. A single spin's transverse coherence decays as $e^{-2\gamma t}$. The dissipator does not respect the scars.

Methods.

- **Exact evolution** of the real Pauli-coefficient vector (dim $= 4^L$, no symmetry is left with random c), $\exp(\mathcal{L}\, dt)$ by a Taylor series summed to round-off, matrix-free for $L \geq 11$. Up to $L = 14$.
- **Diffusive trajectories:** the dephasing as a fluctuating z field, one step

$$e^{-iH dt/2} \prod_j e^{-i\sqrt{\gamma}\,\Delta W_j \sigma^z_j}\, e^{-iH dt/2},$$

  error of the average $O(dt^2)$; 224 trajectories (112 at $L = 20$), $L = 12, \ldots, 20$.
- **Jump trajectories:** since $(\sigma^z_j)^2 = 1$, the effective Hamiltonian is H up to a constant, and a trajectory is exact unitary evolution under H interrupted by σ^z_k kicks, with exponential waiting times of total rate γL and a uniform kicked site. The sampling is exact; from the scar state, the state before the first kick is the analytic product state. 448 trajectories (224 at $L = 18, 20$, $\gamma < 0.002$) at the small γ of Secs. 5–6.
- **Single flip:** one kick σ^z_k on the noiseless scar state at phase $\varphi = ht'$, then evolution with H only (Sec. 6).

Validation: diffusive and jump trajectories agree with the exact ρ at $L = 8$, $\gamma = 0.1$ and 0.01 (2000 trajectories, rms deviation ≈ 1 standard error); at the production sizes see Table 4.

2 How the decay rates are computed

All rates in Secs. 4–6 are obtained the same way, from the time series of one signal $S(t)$ per run: $m_\perp$ for the scar, $s_\perp$ for the x-Néel state, $|s_z|$ for the z-Néel state (Table 1). All three start at $S(0) = 1$. Code: `rates` in `plot_scripts/fit_states.py` and `rate` in `fit_small_gamma.py`.

Envelope. The signal precesses (the scar's $m_\perp$ is smooth, but s_z crosses zero, and small systems show revivals), so the rate is read from the non-increasing envelope

$$E(t) = \max_{t' \geq t} S(t'), \tag{3}$$

computed on the output grid ($dt = 0.1$, or 0.25 for the small-γ jump runs).

The $1/e$ rate Γ_e (main rate). t_e is the first time with $E(t_e) < 1/e$. Between the two output times t_{i-1}, t_i that bracket the crossing it is interpolated linearly in $\ln E$:

$$t_e = t_{i-1} + (t_i - t_{i-1}) \frac{\ln E_{i-1} + 1}{\ln E_{i-1} - \ln E_i}, \qquad \Gamma_e = \frac{1}{t_e}. \tag{4}$$

This is *not* a fit of $A_0 e^{-\Gamma t}$: there is no free amplitude and no fit window. It is the rate of the exponential that starts at the true initial value $S(0) = 1$ and passes through the measured envelope at $1/e$. For a single exponential $e^{-\Gamma t}$ it returns Γ exactly. For a general decay, writing $E(t) = \exp[-\int_0^t \lambda(t')\, dt']$ with the instantaneous rate $\lambda = -d \ln E/dt$,

$$\Gamma_e = \frac{1}{t_e} \int_0^{t_e} \lambda(t)\, dt, \tag{5}$$

the instantaneous rate averaged from $t = 0$ to the $1/e$ time. This matters here because the decays are not single exponentials. The scar decays multi-exponentially, and at small γ it follows $\exp[-\gamma \int \bar{D}]$, whose instantaneous rate grows from 2γ to γD_{sat} (Sec. 6). A fit of $A_0 e^{-\Gamma t}$ with A_0 free would instead depend on the window (Appendix A): with an early fast drop it gives $A_0 < 1$ and the slower late rate. The single-kick prediction of Γ_e uses the same definition, $\gamma \int_0^{t_e} \bar{D} = 1$.

Statistical error (trajectories). With $\sigma_m(t)$ the standard error of the trajectory-averaged signal (from the errors of m_x, m_y or s_x, s_y), the error of the crossing time is $\sigma_t = \sigma_m(t_e)/|dE/dt|$, with the slope taken over ± 4 output steps, so that $\sigma_\Gamma = \Gamma_e^2 \sigma_t$. For an exponential near t_e, $\sigma_\Gamma/\Gamma_e \approx e\, \sigma_m$. The exact evolution has no statistical error.

Tail rate Γ_{tail}. A least-squares straight-line fit of $\ln E$ between the times where E drops below 10^{-2} and 10^{-3}. This one *is* an $A_0 e^{-\Gamma t}$ fit, with A_0 free, on the tail only; it measures the slowest modes (the tower edges for the scar). It is used for exact runs only: trajectory runs reach their statistical floor in that window. Runs stop once the signal stays below $\max(10^{-3}, 3\sigma)$ for 10 time units, or at a fixed $t_{\max}$ (the small-γ runs, $t_{\max} = 3/(\gamma(L-1))$, past t_e).

Thermodynamic rate Γ_∞. For every state and γ, $\Gamma_e(L) = A + B/L$ is fitted (unweighted) on $L \geq 8$ and $\Gamma_\infty = A$. The error is the larger of the fit error and $|A - A_{\text{quad}}|$, with A_{quad} the intercept of a quadratic fit in $1/L$ on $L \geq 6$. Three groups of sizes and γ are handled differently:

- **Néel states with finite-size revivals.** The affected small sizes are left out: x-Néel $L = 6$ at $\gamma \leq 0.05$, z-Néel $L = 6, 8$ at $\gamma \leq 0.02$. z-Néel has no quadratic fit; its error is $|A - \Gamma_e(L_{\max})|$.
- **Scar at $\gamma < 0.01$.** Every size sits at the cap $\Gamma_e \approx \gamma(L-1)$ (Sec. 5), so the $L \geq 8$ fit is not used. Instead:
  - Γ_e of the largest size is a lower bound on Γ_∞, since Γ_e grows with L there.
  - A second fit $A + B/L$ on $L = 14$–20 only gives an estimate. Its error is the larger of the fit error and the change of A when $L = 12$ is added (Table 6).
  - The collapse $\Gamma_e/\gamma = \xi f(L/\xi)$, $\xi = \gamma^{-\nu}$, gives the power law $\Gamma_\infty \propto \gamma^{1-\nu}$.
- **$\gamma = 0$.** It is not fitted. Its Γ_e is the mean over the fitted sizes; the scar never reaches $1/e$, so its Γ_e is 0.

3 Exact results

These hold for any coefficients.

Dark states and steady states. $|\Phi_0\rangle = |\uparrow\cdots\uparrow\rangle$ and $|\Phi_L\rangle = |\downarrow\cdots\downarrow\rangle$ are eigenstates of H and of every σ_j^z, hence dark states of the dissipator. The kernel of $\mathcal{L}$ is spanned by

$$\mathbb{1}, \qquad |\Phi_0\rangle\langle\Phi_0|, \qquad |\Phi_L\rangle\langle\Phi_L|, \tag{6}$$

and ED finds exactly three zero eigenvalues for $L = 4, 5, 6$. The infinite-temperature state is therefore *not* the unique steady state. The coherence $|\Phi_0\rangle\langle\Phi_L|$ is an eigenmode with $\lambda = -2\gamma L - ihL$, which saturates the general bound $-2\gamma L \le \mathrm{Re}\,\lambda \le 0$.

Edge coherences of the tower. Since $\sigma_j^z|\Phi_0\rangle = |\Phi_0\rangle$ and $\sum_j \sigma_j^z = 2S^z_{\mathrm{tot}}$,

$$\mathcal{D}\big(|\Phi_0\rangle\langle\Phi_n|\big) = \gamma\,|\Phi_0\rangle\langle\Phi_n|\Big(\sum_j \sigma_j^z - L\Big) = -2n\gamma\,|\Phi_0\rangle\langle\Phi_n|, \tag{7}$$

and both states are eigenstates of H, so $|\Phi_0\rangle\langle\Phi_n|$ (and likewise $|\Phi_L\rangle\langle\Phi_{L-n}|$) is an exact Liouvillian eigenmode with

$$\lambda = -2n\gamma - ihn \qquad \text{for every } L. \tag{8}$$

The slowest tower modes therefore decay at exactly 2γ, independent of L. In first-order perturbation theory the coherences $|\Phi_n\rangle\langle\Phi_{n+1}|$ decay at $\gamma L(1 - m_n m_{n+1})$ with $m_n = 1 - 2n/L$: 2γ at the edges, $\approx \gamma L$ in the middle of the tower. The scar signal is a binomially weighted sum of these, which makes it multi-exponential.

Initial loss of tower weight. For $\rho = |\psi\rangle\langle\psi|$ with $|\psi\rangle = |-\rangle^{\otimes L}$, $w = \mathrm{Tr}(P_{\mathrm{tower}}\rho)$ and $[P_{\mathrm{tower}}, H] = 0$,

$$\dot w(0) = \gamma\sum_j \big(\|P_{\mathrm{tower}}\sigma_j^z|\psi\rangle\|^2 - 1\big), \qquad P_{\mathrm{tower}}\sigma_j^z|\psi\rangle = \tfrac{2}{L}S^z_{\mathrm{tot}}|\psi\rangle \;\Rightarrow\; \dot w(0) = -\gamma(L-1), \tag{9}$$

using permutation symmetry of the tower and $\langle (S^z_{\mathrm{tot}})^2\rangle = L/4$. A kick keeps a fraction $1/L$ of the state in the tower; the rest is $k \neq 0$ spin waves. Reproduced exactly by the numerics for $L = 4, \ldots, 8$. At late times $w \to (L+1)/2^L$.

Overlaps. $P_{\mathrm{tower}}\sigma_j^\alpha P_{\mathrm{tower}} = \frac{2}{L}S^\alpha_{\mathrm{tot}}P_{\mathrm{tower}}$ for every j, so the staggered operators $O_\pi^\alpha = \sum_j (-1)^j\sigma_j^\alpha$ have zero tower part. The x-Néel signal (Sec. 4) is therefore blind to the tower.

4 Scar against thermal decay up to $L = 20$

4.1 States and signals

state	ρ_0	signal	at $H_I = 0$
scar	$\|-\rangle^{\otimes L}$	$m_\perp = \sqrt{m_x^2 + m_y^2}$	$e^{-2\gamma t}$
x-Néel (thermal reference)	$\|+-+-\cdots\rangle$, even L	$s_\perp = \sqrt{s_x^2 + s_y^2}$	$e^{-2\gamma t}$
z-Néel	$\|\uparrow\downarrow\uparrow\downarrow\cdots\rangle$, even L	$\|s_z\|$	no decay

Table 1: Initial states, $m_\alpha = \frac{1}{L}\sum_j\langle\sigma_j^\alpha\rangle$, $s_\alpha = \frac{1}{L}\sum_j(-1)^j\langle\sigma_j^\alpha\rangle$.

The z-Néel state is a biased reference: it is diagonal in the z basis, so $\mathcal{D}(\rho_0) = 0$, and s_z is invisible to the bath, $\mathcal{D}^\dagger(\sigma_j^z) = 0$. The bath acts only after H_I has rotated weight into σ^x, σ^y: with $H_I = 0$, s_z never decays, and at large γ it is Zeno-protected. The x-Néel $s_\perp$ has the same free-spin baseline $e^{-2\gamma t}$ as the scar's $m_\perp$ and zero tower overlap, so it is the like-for-like thermal reference. Neither Néel state is a dark state of $\mathcal{L}$.

Runs. Rates as in Sec. 2. The scar is computed for $L = 4, \ldots, 12$ and 14 exactly and $L = 16, 18, 20$ with trajectories; the Néel states for even L (exact up to 14 for x-Néel and up to 12 for z-Néel, trajectories $L = 16, 18, 20$ for x-Néel). The γ grid is $0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10$. Where exact and trajectory runs overlap ($L = 12, 14$), Γ_e agrees to 1–2.5% for $\gamma \leq 2$.

4.2 Scar

At $\gamma = 0$ the scar state precesses perfectly, $\langle \sigma_j^x \rangle = -\cos ht$, $\langle \sigma_j^y \rangle = -\sin ht$, on every site and for every L ($\lesssim 10^{-13}$). With dephasing, $m_\perp$ decays faster than the free-spin $e^{-2\gamma t}$ (Fig. 1). Table 2 gives Γ_e/γ.

γ	$L = 8$	10	12	14	16	18	20
0.01	6.73	7.84	8.49	8.88	9.56	10.24	10.61
0.02	6.00	6.77	7.77	8.15	8.34	8.50	8.64
0.05	5.76	6.03	6.24	6.25	6.26	6.28	6.32
0.1	4.36	4.45	4.48	4.47	4.43	4.46	4.51

Table 2: Scar Γ_e/γ (exact for $L \leq 14$, trajectories for $L \geq 16$). $\gamma = 0.2$: 3.0–3.3; $\gamma = 1$: 2.21–2.26; $\gamma = 10$: ≈ 2.0.

For $\gamma \geq 0.05$, Γ_e is L-independent from $L \approx 10$ to 20 (within $\approx 1\%$); at $\gamma = 0.02$ it flattens from $L \approx 16$; at $\gamma = 0.01$ it still grows at $L = 20$. At fixed L and $\gamma \to 0$, $\Gamma_e/\gamma \to L - 1$ (Sec. 5), so a size L shows the thermodynamic rate only if $L - 1 \gg \Gamma_\infty/\gamma$. The extrapolated rates are in Table 3 and Figs. 5, 6. The local exponent $d \ln \Gamma_\infty / d \ln \gamma$ is 0.73 (0.01–0.02), 0.50 (0.02–0.05), 0.44 (0.05–0.1), 0.51 (0.1–0.2), then 0.7–0.9 $\to 1$ at large γ: the lifetime grows as $\gamma^{-1/2}$ over $0.02 \leq \gamma \leq 0.2$. The steeper step at 0.01–0.02 rests on the least converged point: at $\gamma = 0.01$ the fit on $L \geq 14$ gives 0.147 ± 0.009 instead of 0.128 (Sec. 5, Table 6), which brings the exponent there to 0.41. At large γ, $\Gamma_\infty \to 2\gamma$: the dephasing outruns H_I and every spin dephases on its own.

γ	0.01	0.02	0.05	0.1	0.2	0.5	1	2	5	10
scar Γ_∞	0.128(12)	0.213(10)	0.335(14)	0.453(10)	0.647(22)	1.24(2)	2.26(1)	4.22(8)	9.5(5)	19.7(3)
scar Γ_∞/γ	12.8	10.6	6.7	4.5	3.2	2.5	2.26	2.1	1.9	1.97
x-Néel Γ_∞	4.74(36)	4.75(38)	4.78(34)	4.90(13)	4.99(11)	5.32(11)	5.78(8)	7.04(15)	11.3(3)	20.6(3)
z-Néel Γ_∞	5.04(9)	5.03(9)	5.08(19)	4.89(7)	4.72(5)	4.28(2)	3.77(3)	2.91(2)	1.71	0.98

Table 3: Thermodynamic rates. At $\gamma = 0$: scar 0 (perfect revivals), x-Néel 4.76 ± 0.08 (mean Γ_e over $L \geq 8$), z-Néel 4.96 ($L = 10, 12$). z-Néel from $L \leq 12$ only.

4.3 Thermal references

x-Néel. At $\gamma = 0$ it thermalizes with $\Gamma_e \approx 4.8$ for every $L \geq 8$ ($L = 4, 6$ have finite-size revivals and never reach $1/e$ at $\gamma \leq 0.05$). Γ_e is L-independent for $L \geq 8$ at every γ, within $\approx 1\%$ (the $L = 16$ realization lies $\approx 4\%$ lower at every γ, also at $\gamma = 0$: a property of its c). Γ_∞ stays at the closed-system rate for $\gamma \leq 0.1$ and then approaches 2γ (+0.6 at $\gamma = 10$): there is no Zeno slowdown (Fig. 2).

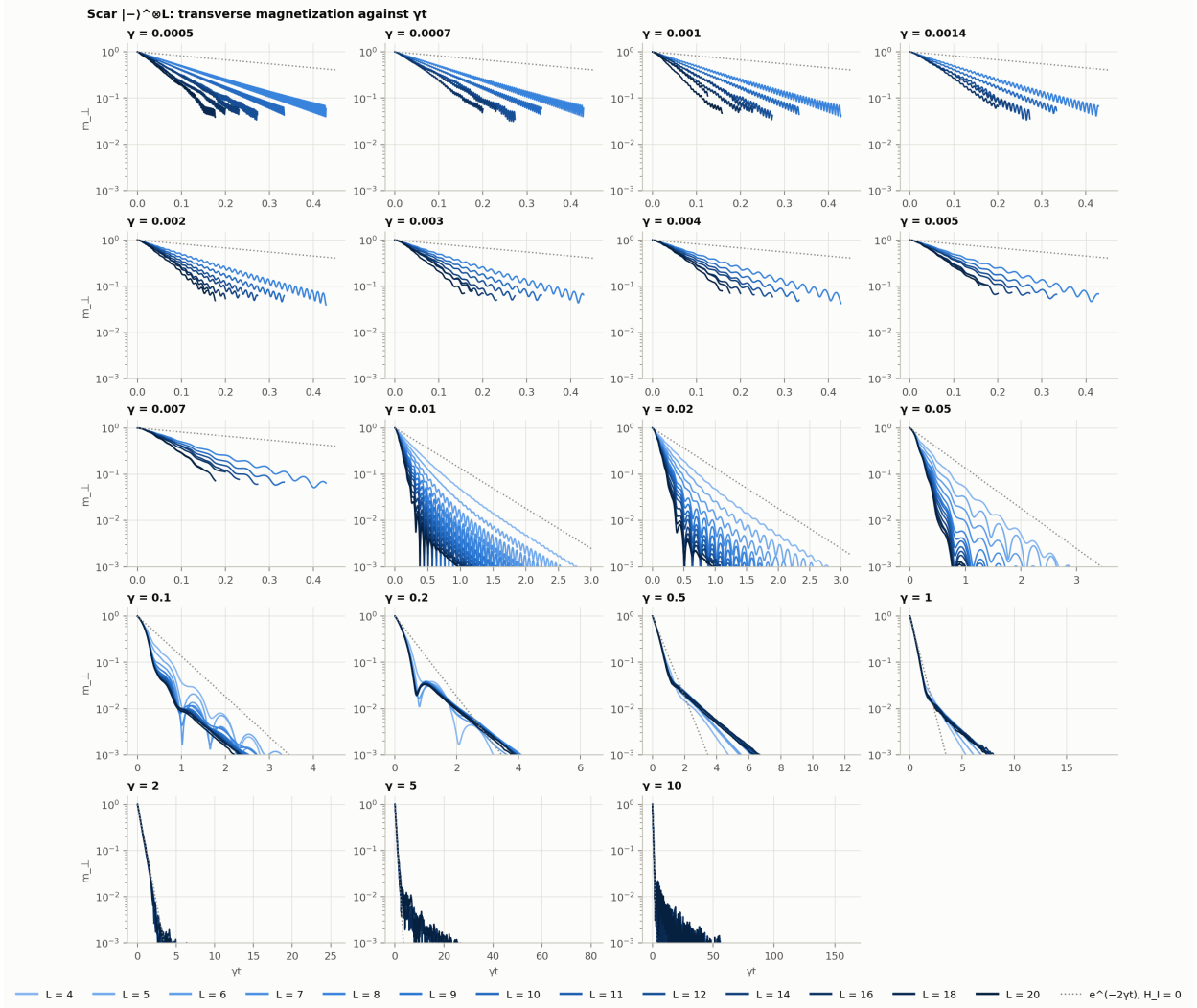


Figure 1: Scar: $m_\perp$ against γt for every γ and L; dotted: $e^{-2\gamma t}$. The panels with $\gamma < 0.01$ are the small-γ runs of Sec. 5, which stop at $t = 3/(\gamma(L-1))$ (`plot_scripts/plot_states.py`).

z-Néel. At $\gamma = 0$, $|s_z|$ decays at $\Gamma_e \approx 5$ for $L \geq 10$ ($L = 4$ has strong revivals). $\Gamma_\infty \approx 5.0$ for $\gamma \leq 0.05$, then decreases, $\propto 1/\gamma$ at large γ (0.98 at $\gamma = 10$): a Zeno slowdown, as expected for a signal the bath cannot see directly (Fig. 3). The number of down spins (Fig. 4) rises at $\gamma = 0$ from $L/2$ to $\approx 0.55L$ ($m_z \approx -0.1$, the thermal value fixed by the energy since H contains $hS^z_{\rm tot}$), and with dephasing overshoots and relaxes back to $L/2$ at ≈ 1.5–2.3γ, L-independent for $L \geq 8$.

Scar against thermal. $\Gamma_\infty(\text{scar}) < \Gamma_\infty(x\text{-Néel})$ at every γ, by a factor 37 ($\gamma = 0.01$), 22 (0.02), 14 (0.05), 11 (0.1), 7.7 (0.2), 4.3 (0.5), 2.6 (1), 1.7 (2), 1.05 (10). The scar rate goes to 0 as $\gamma \to 0$; the thermal rate stays at ≈ 4.8 (Fig. 6). This is the expected contrast of the Introduction of the project: scar revivals are regular in the weak-dissipation limit, the thermal signal is not. The z-Néel rate crosses the scar rate at $\gamma \approx 1.5$, an artefact of its Zeno protection.

5 Small γ: the finite-size cap

To reach the regime $\gamma \to 0$, the scar was run at $\gamma = 0.0005$, 0.0007, 0.001, 0.0014, 0.002, 0.003, 0.004, 0.005, 0.007 with jump trajectories for $L = 12, \ldots, 20$ (without 0.0007, 0.0014 at $L = 18, 20$) and exactly for $L = 8, 10$ (every γ) and 12 ($\gamma \geq 0.002$), up to $t = 3/(\gamma(L-1))$. The traj-seeds of one (L, γ) are pooled, weighted by their trajectory counts.

Cross-checks. The jump trajectories have a large per-trajectory spread (≈ 0.3 at $m_\perp \approx 0.5$, against ≈ 0.08 for diffusive trajectories at $L = 12$): a trajectory is either undamaged or carries a damaged region. With 448 trajectories the statistical error of Γ_e is 1.5–6%, up to 7.5% at $L = 20$ with 224. All γ of one L use the same random numbers, so the kick sequences coincide up to the time scale $1/\gamma$ and the errors are correlated across γ. Table 4 compares jump runs with the exact evolution and with diffusive trajectories: the differences are within $\approx 2\sigma$ (Fig. 7); exact and diffusive agree with each other to 0.3%.

L	γ	reference	rms z	$\Gamma_e^{\rm jump}/\Gamma_e^{\rm ref} - 1$
12	0.002	exact	0.74	+0.8%
12	0.003	exact	1.15	−7.1%
12	0.004	exact	1.04	+2.2%
12	0.005	exact	1.83	−10.6%
12	0.007	exact	1.22	−5.1%
12	0.002	diffusive	0.71	+0.4%
12	0.005	diffusive	1.72	−10.4%
14	0.002	diffusive	0.91	+0.3%
14	0.005	diffusive	0.87	−6.4%

Table 4: Jump trajectories against the references: rms over t of $(m_\perp^{\rm jump} - m_\perp^{\rm ref})/\sigma$ while $m_\perp^{\rm ref} > 0.05$, and the difference of Γ_e (statistical error of the jump Γ_e: 4–6%).

Rates. Table 5 and Fig. 8; in Fig. 6 the largest size gives lower bounds on Γ_∞ (0.0085 at $\gamma = 0.0005$, 0.030 at 0.002, 0.084 at 0.007), all below the collapse power law and above 2γ. At $L \leq 10$, $\Gamma_e/\gamma \approx L - 1$ for every $\gamma \leq 0.003$ (6.8–7.0 at $L = 8$, 8.9 at $L = 10$); at $L = 12$–20 the smallest γ reach $L - 1$ within the errors (11.5 at $L = 12$, 14.6 at $L = 16$, 16.9–19.0 at $L = 20$). At fixed L, Γ_e/γ decreases with γ towards Γ_∞/γ. The cap equals the initial loss rate of the tower weight, Eq. (9): at $\gamma \to 0$ with L fixed, the local envelope decays like the global tower weight, because the damage of a kick has time to spread over the whole chain before the next kick (Sec. 6).

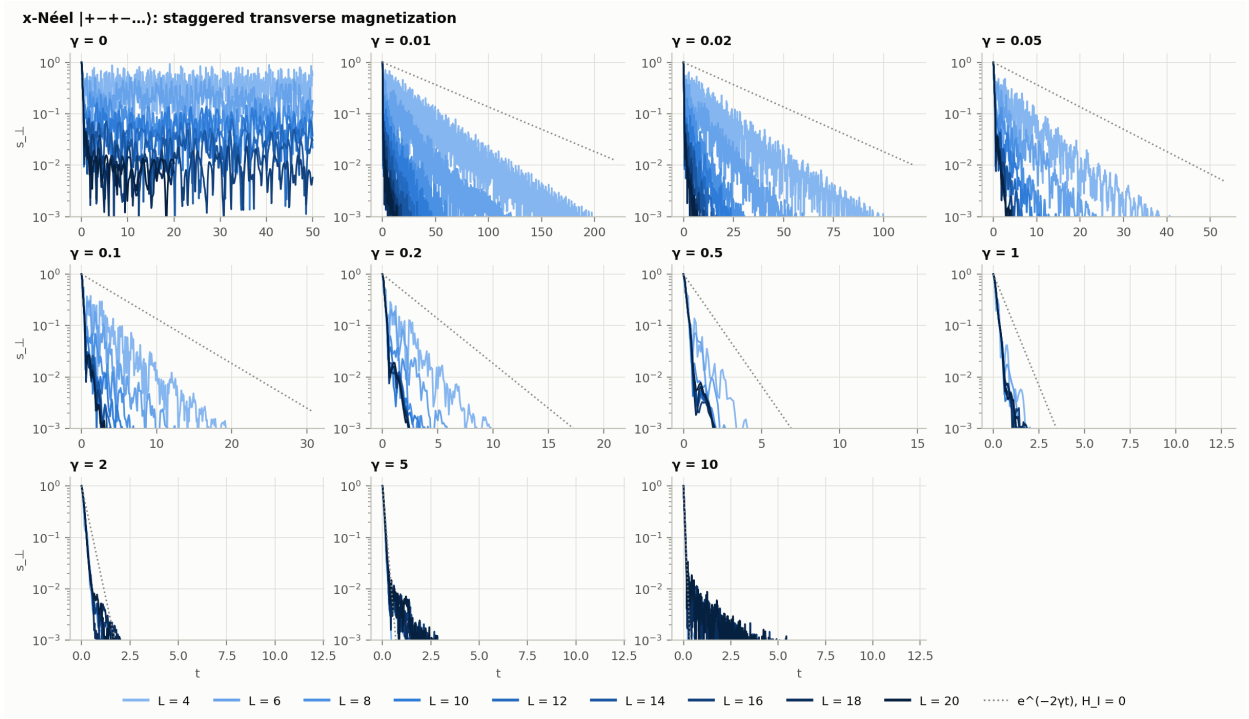


Figure 2: x-Néel: $s_\perp$ against t; dotted: $e^{-2\gamma t}$ (`plot_states.py`).

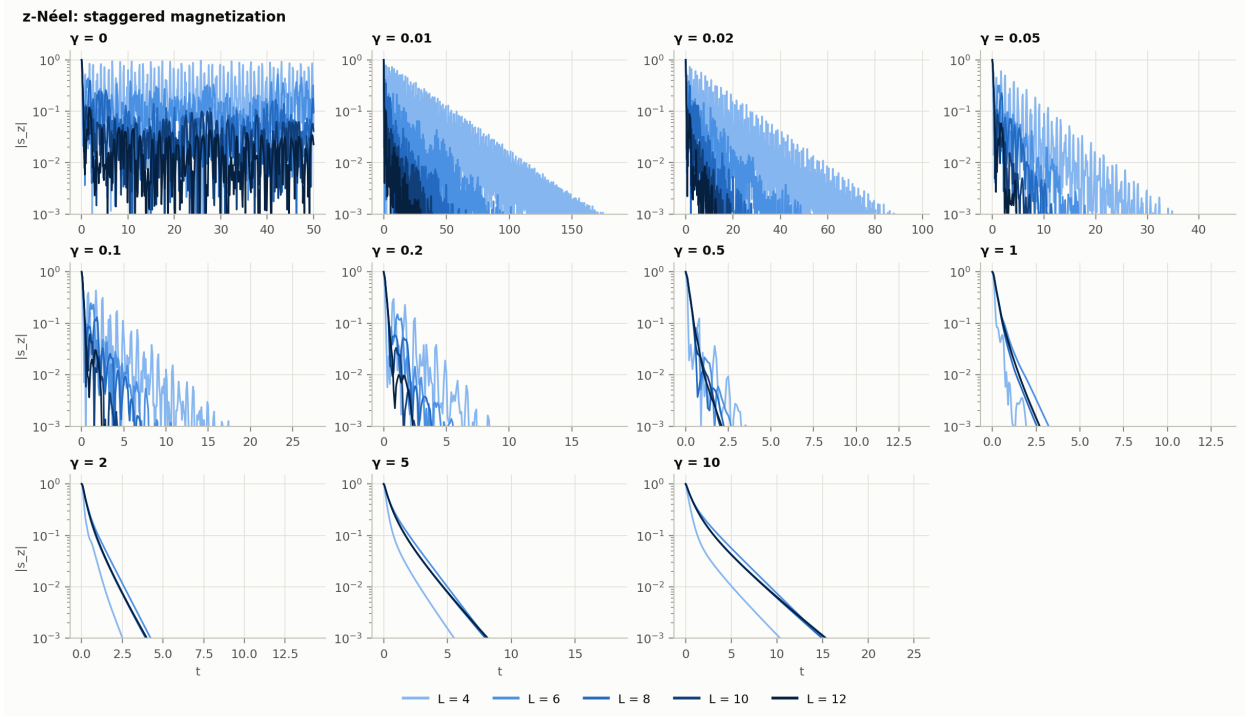


Figure 3: z-Néel: $|s_z|$ against t (`plot_states.py`).

γ	$L=8$	10	12	14	16	18	20
0.0005	6.82	8.89	11.46	11.82	14.60	14.65	16.92
0.0007	6.93	8.82	10.95	11.45	13.49	–	–
0.001	6.98	8.92	11.55	11.48	14.10	14.64	19.04
0.0014	6.81	8.86	10.71	10.69	12.71	–	–
0.002	6.81	8.93	10.24	11.67	13.25	13.34	15.14
0.003	6.85	8.91	10.06	11.10	11.94	13.09	15.30
0.004	6.88	8.37	9.89	10.56	11.73	11.84	13.21
0.005	6.80	8.31	9.31	9.92	11.39	11.66	12.56
0.007	7.17	7.99	8.75	9.23	10.08	11.92	12.01
0.01	6.73	7.84	8.49	8.88	9.56	10.24	10.61
$L-1$	7	9	11	13	15	17	19

Table 5: Scar Γ_e/γ at small γ: exact for $L \le 12$ ($L = 12$ only $\gamma \ge 0.002$), jump trajectories otherwise; $\gamma = 0.01$ from Table 2.

Collapse. If the finite-size effects are set by one length $\xi(\gamma) = \gamma^{-\nu}$, then $\Gamma_e/\gamma = \xi f(L/\xi)$ with $f(x) \approx x$ at small x (the cap) and $f \to$ const at large x, so that $\Gamma_\infty \propto \gamma^{1-\nu}$. Minimizing the spread of $(\Gamma_e/\gamma)\gamma^\nu$ at common $L\gamma^\nu$ ($\gamma \le 0.05$, $L \ge 8$) gives $\nu = 0.33$, with a cost within a factor 2 of the minimum for $\nu = 0.24$–0.48 (the cost at $\nu = 0.5$ is 2.2× the minimum). Ballistic spreading of the damage ($\xi \sim vt$, $\nu = 1/2$, lifetime $\propto \gamma^{-1/2}$) and diffusive spreading ($\nu = 1/3$, lifetime $\propto \gamma^{-2/3}$) are both within this range: $L \le 20$ does not separate them.

Extrapolation from the large sizes. The fit $A + B/L$ on $L \ge 8$ (Sec. 4) includes sizes that are still at the cap when γ is small. A second fit uses only $L = 14, 16, 18, 20$. Its error is the larger of the fit error and the change of A when $L = 12$ is added. It needs at least three sizes, so $\gamma = 0.0007$ and 0.0014 (only up to $L = 16$) are left out. Results are in Table 6 and Fig. 9. They appear as the $L = \infty$ points of Fig. 8 and as open squares in Fig. 6.

- At $\gamma \ge 0.05$ the two fits agree.
- At $\gamma = 0.01$ and 0.02 they differ by 15% and 8%, with opposite signs: the $L \ge 8$ value at $\gamma = 0.01$ (0.128) is pulled down by the capped small sizes.
- At $\gamma < 0.01$ the extrapolated $A/\gamma = 18$–34 lies 40–80% above $\Gamma_e(L = 20)/\gamma$. A moves by up to 25% when $L = 12$ is added, so these values are estimates, not converged limits. They sit inside the band of the collapse power law.

A power law $A \propto \gamma^p$ fitted to all ten values ($0.0005 \le \gamma \le 0.05$) gives $p = 0.58 \pm 0.01$ (weighted) or 0.66 ± 0.04 (unweighted), i.e. $\nu = 1 - p = 0.34$–0.42, consistent with the collapse. Restricted to $\gamma \le 0.007$ it gives $p = 0.80 \pm 0.06$. That steepening is expected from the fit itself: at smaller γ the sizes are closer to the cap, so the extrapolation underestimates Γ_∞ more.

γ	0.0005	0.001	0.002	0.003	0.004	0.005	0.007	0.01	0.02	0.05
A/γ, $L \ge 14$	27.4	33.8	22.1	24.0	18.5	18.3	19.4	14.8	9.8	6.5
error (%)	11	19	9	11	8	6	10	6	2	1
A/γ, $L \ge 12$	24.3	27.4	21.6	21.6	17.5	17.3	17.4	13.8	9.9	6.4
$\Gamma_e(L = 20)/\gamma$	16.9	19.0	15.1	15.3	13.2	12.6	12.0	10.6	8.6	6.3
Γ_∞/γ, $L \ge 8$ (Table 3)	–	–	–	–	–	–	–	12.8	10.6	6.7

Table 6: Scar: $A + B/L$ on the large sizes ($L = 14, \ldots, 20$), with the check on $L \ge 12$ (`fit_small_gamma.py`, `states_inverse_L_large.csv`).

6 Single-kick test

Model. Take a kick σ_k^z at time t' on the otherwise noiseless scar state, and evolve it with H for a time τ. With $c_j(\tau) = (\langle\sigma_j^x\rangle + i\langle\sigma_j^y\rangle)$ divided by its noiseless value at $t'+\tau$, define the damage

$$D(\tau) = \sum_j \big[1 - \mathrm{Re}\, c_j(\tau)\big], \qquad D(0) = 2, \tag{10}$$

and its average $\bar{D}(\tau)$ over 16 phases $\varphi = ht'$ and the kicked sites. If the damage of different kicks is independent and multiplies, each kick multiplies $m_\perp$ by $1 - D/L$. With kicks at total rate γL, the Poisson average gives

$$m_\perp(t) \approx \exp\Big[-\gamma L \int_0^t \frac{\bar{D}(\tau)}{L}\, d\tau\Big] = \exp\Big[-\gamma \int_0^t \bar{D}(\tau)\, d\tau\Big], \tag{11}$$

which reproduces the free-spin $e^{-2\gamma t}$ for $D \equiv 2$. The predicted Γ_e solves $\gamma \int_0^{t_e} \bar{D} = 1$, with no free parameter. Runs: all sites for $L = 12, \ldots, 20$ up to $\tau = 25$, and four sites $(0, L/4, L/2, 3L/4)$ for $L = 22, 24$ up to $\tau = 8$. Beyond $\tau = 25$, $\bar{D}$ is continued with its mean over $20 \le \tau \le 25$.

Damage. Fig. 10. $\bar{D}$ jumps from 2 to ≈ 5.9 within $\tau = 1$ for every L, then grows, and saturates with slow oscillations (period ≈ 6). At $\tau = 4$: $9.10, 9.99, 10.66, 11.02, 11.35, 11.49, 11.60$ for $L = 12, 14, \ldots, 24$; at $\tau = 8$: $9.73, 11.02, 12.36, 13.44, 14.62, 15.54, 16.32$. The saturation value is $11.0, 12.4, 14.0, 15.4, 17.1$ for $L = 12, \ldots, 20$, i.e. $\approx L - 1$ at $L = 12$ and still 0.6–1.9 below $L - 1$ at $\tau = 25$ for the larger sizes. The size dependence sets in at $\tau \approx 3$ ($L = 12$) to $\tau \approx 6$ ($L = 22$ against 24). For $L = 22, 24$, $\bar{D}$ grows by ≈ 1.0–1.2 per unit of τ between $\tau = 4$ and 8, roughly linearly (only four kicked sites are sampled; the spread over the 64 runs is ≈ 1.2 at $\tau = 8$).

Finite L. The prediction (11) follows the whole measured time series (Fig. 11). Over all 55 points ($L = 12, \ldots, 20$, $\gamma \le 0.05$), the ratio of measured to predicted Γ_e is 0.90–1.19, mostly 0.95–1.05 (Fig. 12, left). At $\gamma < 0.01$ this is within the statistical errors. At $\gamma = 0.01$–0.05, where the errors are below 1%, the ratio is systematically 0.95–1.00: the independent-kick formula counts overlapping damage twice, so it predicts a slightly faster decay. So at small γ the scar decay is fully set by isolated kicks, and the finite-size cap $\Gamma_e/\gamma \to L - 1$ is the saturated damage $D_{\mathrm{sat}} \approx L - 1$.

Thermodynamic limit. Fig. 12, right: Γ_∞ against the prediction from $\bar{D}$ at $L = 24$ ($\tau \le 8$, so $\gamma \ge 0.02$). The two agree at $\gamma = 0.05$ (0.332 against 0.335 ± 0.014). Above that, kicks overlap within the damage time, and the prediction is too fast: 0.53 against 0.45 at $\gamma = 0.1$, 0.84 against 0.65 at 0.2, 1.54 against 1.24 at 0.5. At $\gamma \ge 2$ both go to 2γ. At $\gamma = 0.02$ the prediction (0.181) is 15% below the $L \ge 8$ extrapolation (0.213 ± 0.010), but only 7% below the fit on $L \ge 14$ (0.196 ± 0.003, Table 6), and close to the largest measured size ($\Gamma_e(L = 20) = 0.173$). The remaining gap is either $\bar{D}$ still growing beyond $L = 24$ at $\tau \approx 5$, or the extrapolation.

Consequence. In the thermodynamic limit at small γ, Eq. (11) makes the lifetime a property of $\bar{D}_\infty(\tau)$ alone. Linear growth $\bar{D} \simeq 2v\tau$ (ballistic spreading of the damage) gives $\gamma v t_e^2 = 1$, i.e. $\Gamma_\infty = (\gamma v)^{1/2}$. Diffusive growth $\bar{D} \propto \tau^{1/2}$ gives $\Gamma_\infty \propto \gamma^{2/3}$. If the damage stays bounded, $\Gamma_\infty \propto \gamma$. The measured Γ_∞ ($\Gamma_\infty^2/\gamma \approx 2.0$–$2.3$ for $0.02 \le \gamma \le 0.1$) and the roughly linear growth of $\bar{D}$ at $L = 22, 24$ for $4 \le \tau \le 8$ favour the ballistic case, but the collapse admits both. Deciding needs $\bar{D}$ at longer τ in larger systems.

7 Summary and open points

- The scar transverse magnetization decays at an L-independent rate once $L - 1 \gg \Gamma_\infty/\gamma$. Its thermodynamic rate vanishes as $\gamma \to 0$, roughly as $\gamma^{1/2}$ for $0.02 \le \gamma \le 0.2$, and tends to 2γ

at large γ. The thermal x-Néel signal decays at the closed-system rate ≈ 4.8 for $\gamma \leq 0.1$, so the two differ by a factor 37 at $\gamma = 0.01$. The z-Néel state is Zeno-protected and not a fair reference.

- At fixed L and $\gamma \to 0$, $\Gamma_e/\gamma \to L-1$, the loss rate of the tower weight: the local signal then behaves like a global one.
- Extrapolating only the large sizes ($L = 14$–20) gives estimates of Γ_∞ down to $\gamma = 0.0005$ ($\Gamma_\infty/\gamma \approx 18$–$34$ for $\gamma < 0.01$, not converged), consistent with $\Gamma_\infty \propto \gamma^{0.58}$–$\gamma^{0.66}$ over $0.0005 \leq \gamma \leq 0.05$ and with the collapse.
- A single kick evolved with H predicts the scar decay at every $L \leq 20$ and $\gamma \leq 0.05$ to within $\approx 5\%$ without parameters. The small-γ lifetime is therefore the integral of the single-kick damage.
- Open: the growth law of $\bar{D}_\infty(\tau)$ (ballistic or diffusive) at $\tau > 8$, which fixes the exponent of Γ_∞ as $\gamma \to 0$; one disorder realization per L; the $\gamma = 0.01$–0.02 extrapolation, which is the least converged.

A Earlier results with the previous coefficients ($L \leq 10$)

The results of this appendix (14 October 2026) were obtained with $h = 0.5$, $a = (0.73, 0.41, 0.29)$, $b = (0.58, -0.37, 0.66)$, and only c_{xz}, c_{zy}, c_{yx} nonzero with baseline values $0.31, -0.44, 0.27$ (random factors in $[0.5, 1.5]$ where stated), with exact evolution and full Liouvillian ED for $L \leq 10$, for the scar state and the probes below (the tilted cone initial states studied then are left out). Rates are fits of $A_0 e^{-\Gamma t}$ with A_0 free in fixed windows, not the $1/e$ rates of the main text (Sec. 2).

A.1 Scar oscillations and their decay

Decay against L. The amplitude $m_\perp$ decays faster for larger L (Fig. 13). For the translation-invariant model, fitting $e^{-\Gamma t}$ while $m_\perp > 0.05$ gives the rates in Table 7, well described by $\Gamma \approx 0.63\,\gamma L^{1.13}$. Breaking translation with random c changes them by less than 2%. With the main-text results this growth is identified as the finite-size cap of Sec. 5.

L	4	5	6	7	8	9	10
Γ/γ, uniform c	3.06	3.80	4.66	5.75	6.68	7.61	8.39
Γ/γ, random c	3.06	3.89	4.68	5.81	6.69	–	–

Table 7: Scar decay rate at $\gamma = 0.01$, fit window $m_\perp > 0.05$ (previous coefficients).

A.2 Scar versus non-scar probes

Three probes, evolved with the same Lindbladian (random c, $L = 4, \ldots, 8$, $\gamma = 0$ and 0.01):

- **P1, scar:** $m_\perp(t)$ from $\rho_0 = |-\rangle\langle -|^{\otimes L}$.
- **P2, energy:** $C(t) = \mathrm{Tr}[\tilde{H} e^{\mathcal{L}^\dagger t}(\tilde{H})]/\,\mathrm{Tr}\,\tilde{H}^2$ with $\tilde{H} = H - hS^z_{\mathrm{tot}} P_{\mathrm{tower}} = (1 - P_{\mathrm{tower}})H(1 - P_{\mathrm{tower}})$, conserved at $\gamma = 0$ and without tower component.
- **P3, thermal:** $|C^+(t)|$ for $O^+ = \sum_j (-1)^j \sigma^+_j$ ($k = \pi$, even L), built from separate evolutions of O^x_π and O^y_π so that the precession in h drops out. At $\gamma = 0$ it thermalizes by $t \approx 1$ and then sits at a finite-size floor ($0.17, 0.05, 0.015$ for $L = 4, 6, 8$). Its dephasing rate is $\Delta\Gamma$, the slope of $-\ln(|C^+_\gamma|/|C^+_0|)$ on a common short window. $\mathrm{Tr}(O_k H) = 0$ exactly for $k \neq 0$ (the single-site σ^x content of H is uniform).

Scaling with L. Table 8 and Fig. 15. The scar rate grows close to linearly in L; the non-scar rates stay near the single-spin value 2γ with a weak (energy) or no (thermal) L-dependence.

probe (Γ/γ)	$L=4$	5	6	7	8	power law
P1 scar, window $t \in [5, 60]$	3.22	4.01	4.69	5.67	6.51	$L^{1.01\pm0.03}$
P1 scar, window $t \in [5, m_\perp = 10^{-3}]$	2.61	2.94	3.28	3.87	4.77	$L^{0.84\pm0.12}$
P2 energy $\tilde{H}$	1.83	1.84	2.00	2.22	2.32	$L^{0.37\pm0.07}$
P3 thermal $\Delta\Gamma$, $t \in [0, 0.5]$	1.43	–	1.59	–	1.61	$L^{0.18\pm0.06}$
P3 thermal $\Delta\Gamma$, $t \in [0, 1]$	2.25	–	2.25	–	1.85	$L^{-0.27\pm0.19}$

Table 8: Decay rates at $\gamma = 0.01$ (random c, previous coefficients) and power-law fits $\Gamma = aL^b$.

Caveats.

- The scar decay is multi-exponential (Fig. 14, left): mid-tower coherences ($\sim \gamma L$) dominate early, the exact 2γ edge modes late. Γ_{scar} and its L-exponent depend on the fit window.
- $L = 4, \ldots, 8$ is a factor of two in L: power law and exponential fit about equally well; the exponents are descriptive.
- For P3, a window that ends at twice the noise floor lengthens with L and produces a spurious growth of $\Delta\Gamma$ $(1.43, 2.25, 4.39)$; only common windows are meaningful.
- The predicted initial rate of P2, $2\gamma \sum_P n_{xy} h_P^2 / \sum_P h_P^2$, is 2.7–2.9 γ; the long-time decay is slower (1.8–2.3 γ).

A.3 Liouvillian spectrum

Full ED at $\gamma = 0.01$ for $L = 4, 5, 6$ (Fig. 16). Each right eigenvector R is split into tower ($\|P_{\text{tower}} R P_{\text{tower}}\|$), mixed and bulk weights. Findings:

- three zero modes (Sec. 3), and $\operatorname{Re}\lambda \geq -2\gamma L$ saturated by $|\Phi_0\rangle\langle\Phi_L|$;
- the bulk is centred at $-\operatorname{Re}\lambda \approx \gamma L$ (the mean dephasing rate of a Pauli string);
- the slowest tower modes sit at exactly 2γ (edge coherences); the next ones at 4γ for $L = 5, 6$ (2.94γ at $L = 4$);
- the slowest bulk mode is at 1.67γ, 1.66γ, 1.88γ for $L = 4, 5, 6$, comparable to the P2 rate;
- modes with tower weight > 0.9: 22 of 25, 32 of 36, 38 of 49: at $\gamma = 0.01$ some tower coherences already hybridize with the bulk.

Checks: at $\gamma = 0$ the dense spectrum equals $-i(E_m - E_n)$ to 5×10^{-14} ($L = 4, 5$); $\sum \lambda = \operatorname{Tr}\mathcal{L} = -\gamma L 4^L$; $L + 1$ eigenstates of H in the tower at energies $h(L/2 - n)$.

A.4 Dependence on γ at fixed $L = 6$

Eleven values $\gamma = 0.002, \ldots, 6$, rates fitted on a window fixed in $\gamma t \in [0.05, 0.6]$ (the window $[5, 60]$ at $\gamma = 0.01$).

Three regimes for the scar (Fig. 17): weak dephasing, $\gamma \leq 0.05$, with $\Gamma \propto \gamma^{0.984\pm0.002}$ at $\approx 4.7\gamma$ (the cap $\approx L - 1$ of the main text); a crossover around $\gamma \approx 0.1$, where $\gamma L \approx h$, with a maximum 5.7γ; strong dephasing, $\gamma \gtrsim 1$, with $\Gamma \to 2\gamma$. The non-scar rates are also linear in γ for weak dephasing ($\gamma^{0.977}$ for P2, $\gamma^{1.001}$ for P3). At strong dephasing the energy probe slows down (1.27γ at $\gamma = 6$).

γ	0.002	0.005	0.01	0.02	0.05	0.1	0.2	0.5	1	3	6
P1 scar	4.80	4.78	4.69	4.61	4.59	5.67	3.75	2.47	2.14	2.02	2.00
P2 energy	2.44	2.42	2.39	2.36	2.26	2.09	1.93	1.62	1.42	1.29	1.27
P3 $\Delta\Gamma$	1.60	1.60	1.60	1.60	1.60	1.61	1.62	1.66	1.72	1.88	2.28

Table 9: Γ/γ at $L = 6$ (random c, previous coefficients).

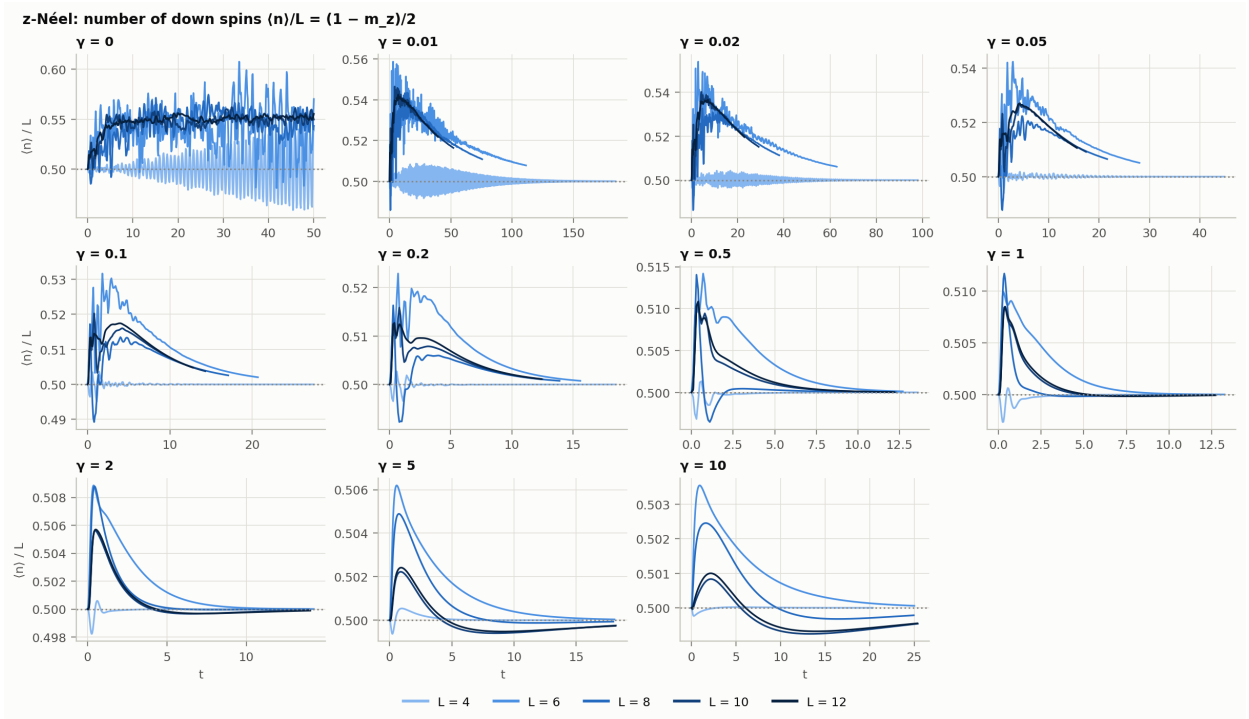


Figure 4: z-Néel: $\langle n\rangle/L = (1 - m_z)/2$; the runs stop on $|s_z|$, so $\langle n\rangle$ is not fully relaxed at small γ (`plot_states.py`).

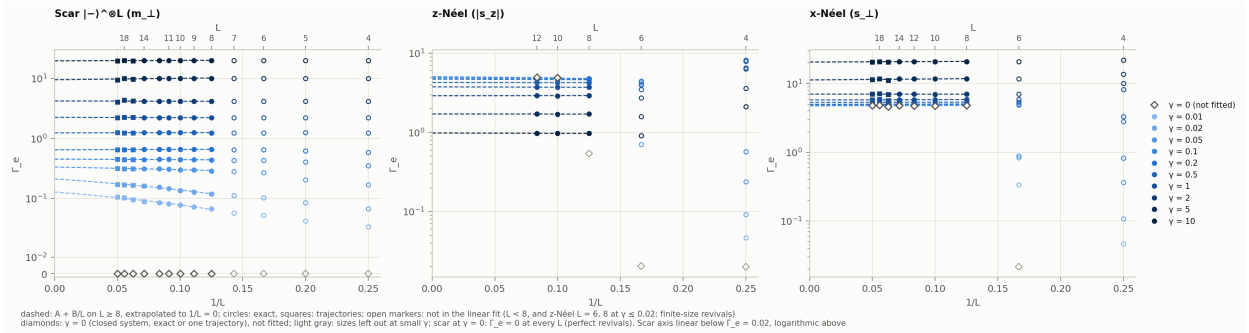


Figure 5: Γ_e against $1/L$ with the fits $A + B/L$ on $L \geq 8$; circles exact, squares trajectories; diamonds $\gamma = 0$, not fitted (`fit_states_inverse_L.py`).

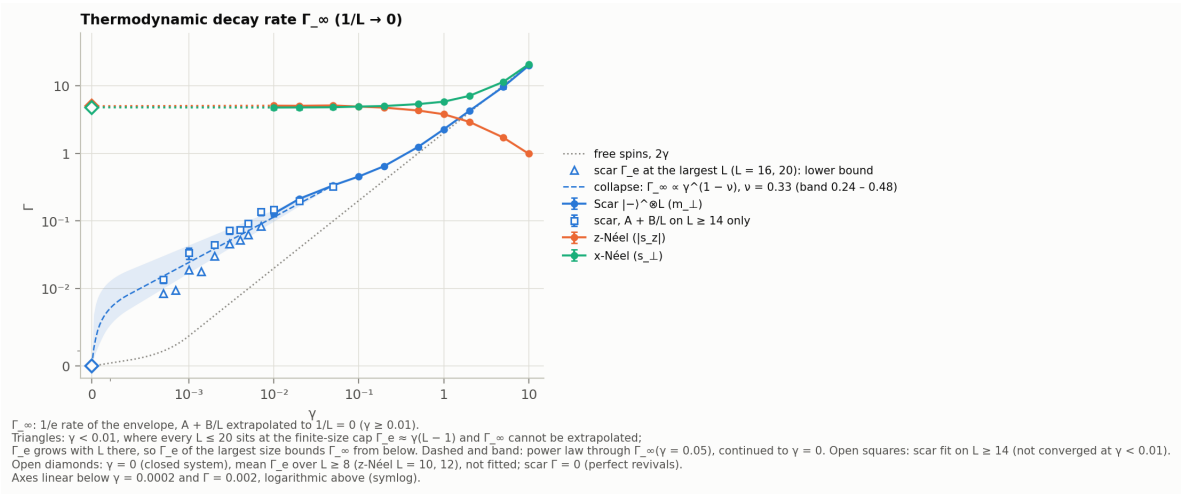


Figure 6: Γ_∞ against γ for the three states (symlog axes; open diamonds: $\gamma = 0$). Scar at $\gamma < 0.01$ (Sec. 5): every $L \leq 20$ sits at the finite-size cap, so Γ_∞ is not extrapolated; triangles are Γ_e of the largest size ($L = 20$; 16 at $\gamma = 0.0007, 0.0014$), lower bounds on Γ_∞ since Γ_e grows with L there. Open squares: $A + B/L$ on $L \geq 14$ only (Table 6; not converged at $\gamma < 0.01$). Dashed line and band: $\Gamma_\infty \propto \gamma^{1-\nu}$ through $\Gamma_\infty(0.05)$ with the collapse exponent $\nu = 0.33$ (0.24–0.48), continued to $\Gamma_\infty = 0$ at $\gamma = 0$ (`fit_states_inverse_L.py`, `fit_small_gamma.py`).

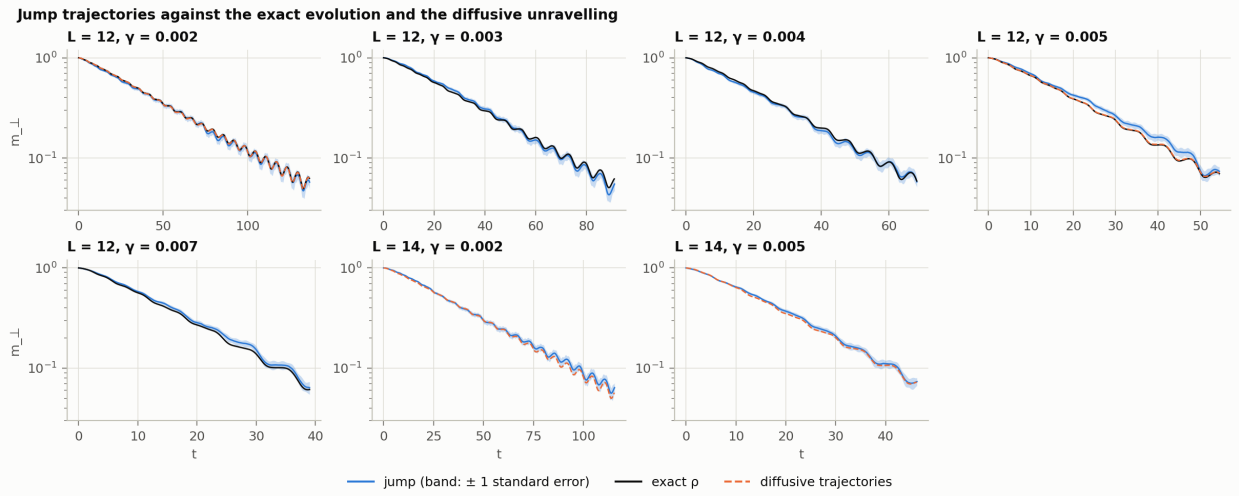


Figure 7: $m_\perp$ of jump trajectories (band: ± 1 standard error) against exact and diffusive runs (`fit_small_gamma.py`).

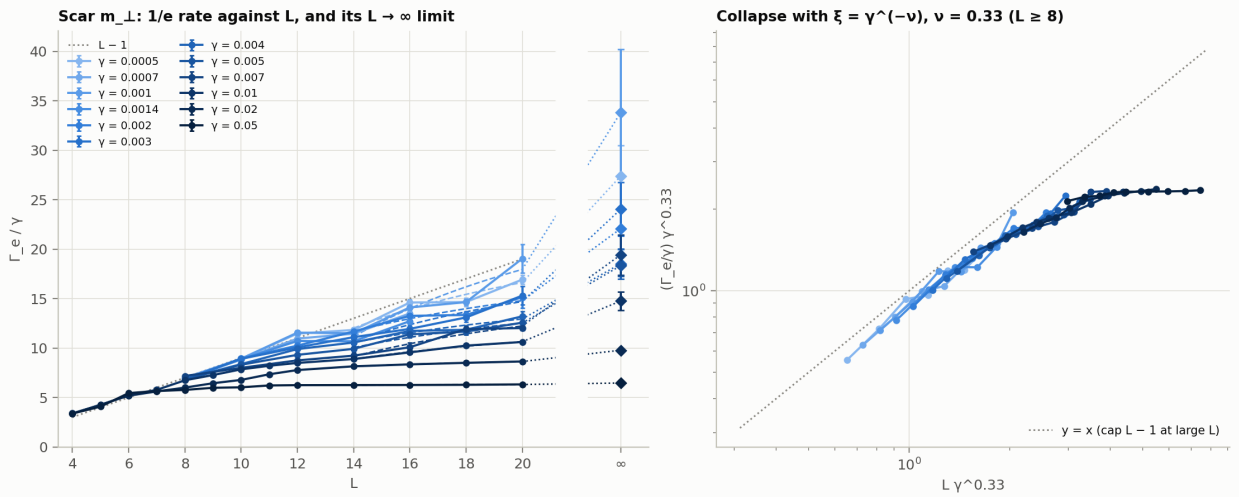


Figure 8: Left: Γ_e/γ against L, one line per γ; dotted: $L-1$; dashed: $A+B/L$ fitted on $L \geq 14$, and diamonds at $L=\infty$: its intercept A/γ (Table 6; not converged at $\gamma < 0.01$). Right: collapse with $\nu = 0.33$ (`fit_small_gamma.py`).

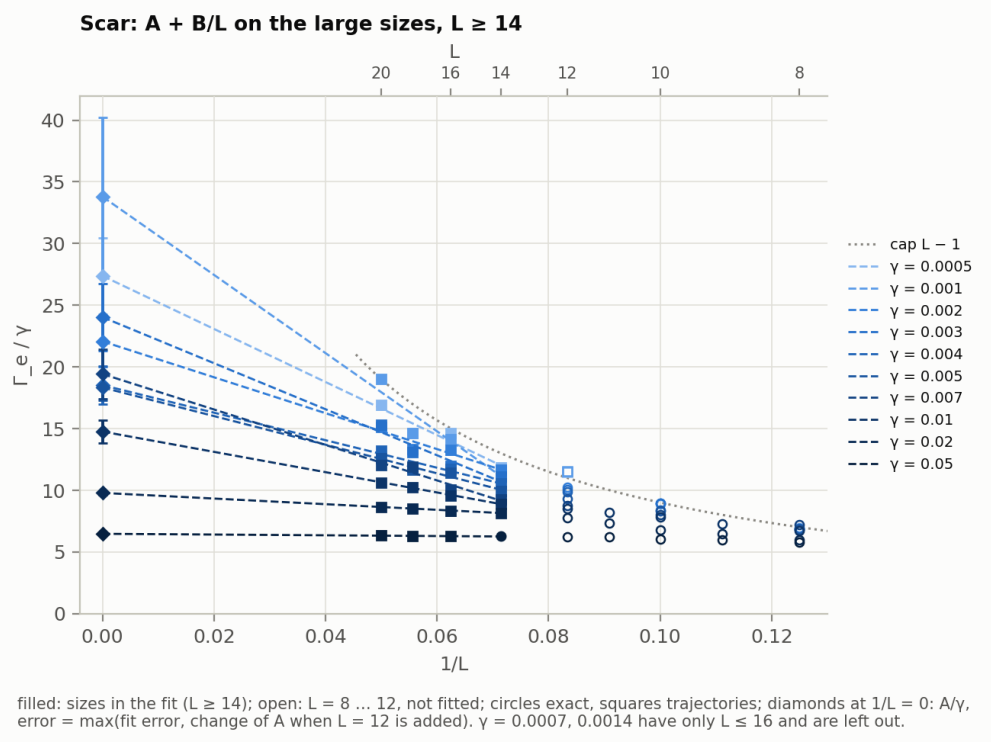


Figure 9: Scar Γ_e/γ against $1/L$ for $\gamma \leq 0.05$, with $A+B/L$ fitted on $L \geq 14$ (filled) and extrapolated to $1/L = 0$ (diamonds); open symbols: $L \leq 12$, not fitted; dotted: the cap $L-1$ (`fit_small_gamma.py`).

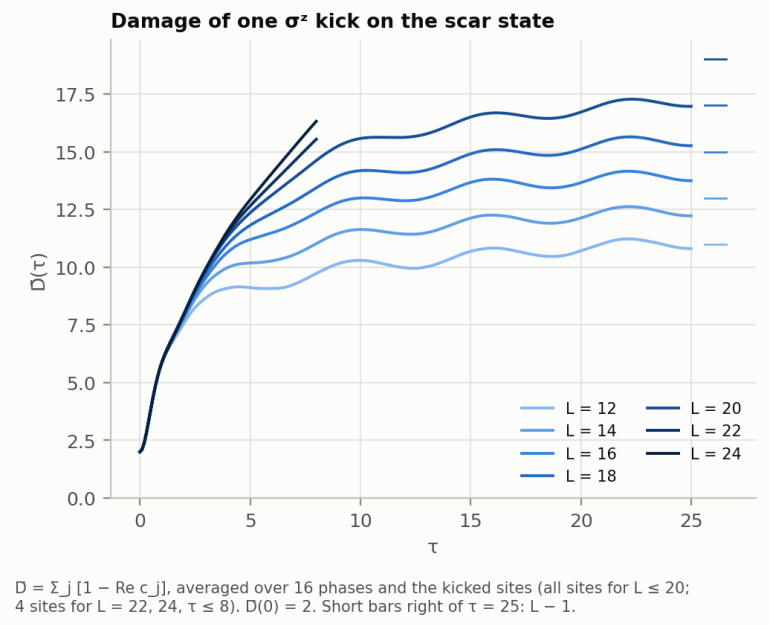


Figure 10: Damage $\bar{D}(\tau)$ of one kick; bars right of $\tau = 25$: $L - 1$ (`single_kick.py`).

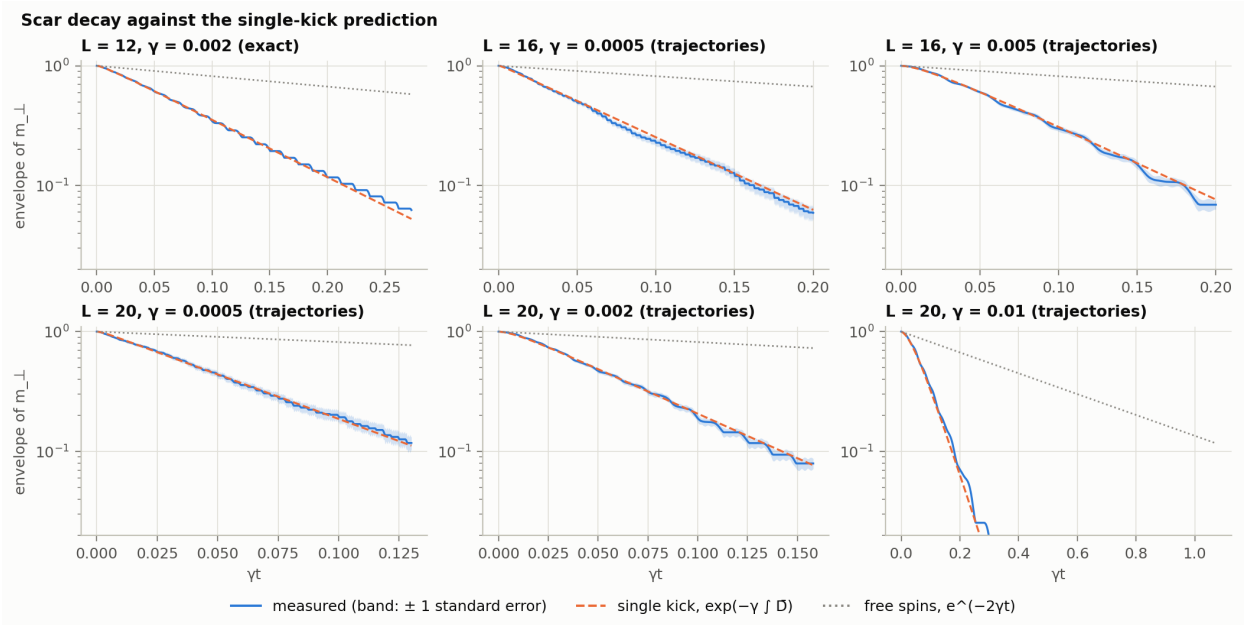


Figure 11: Envelope of $m_\perp$ (band: ± 1 standard error) against the single-kick prediction (11) and $e^{-2\gamma t}$ (`single_kick.py`).

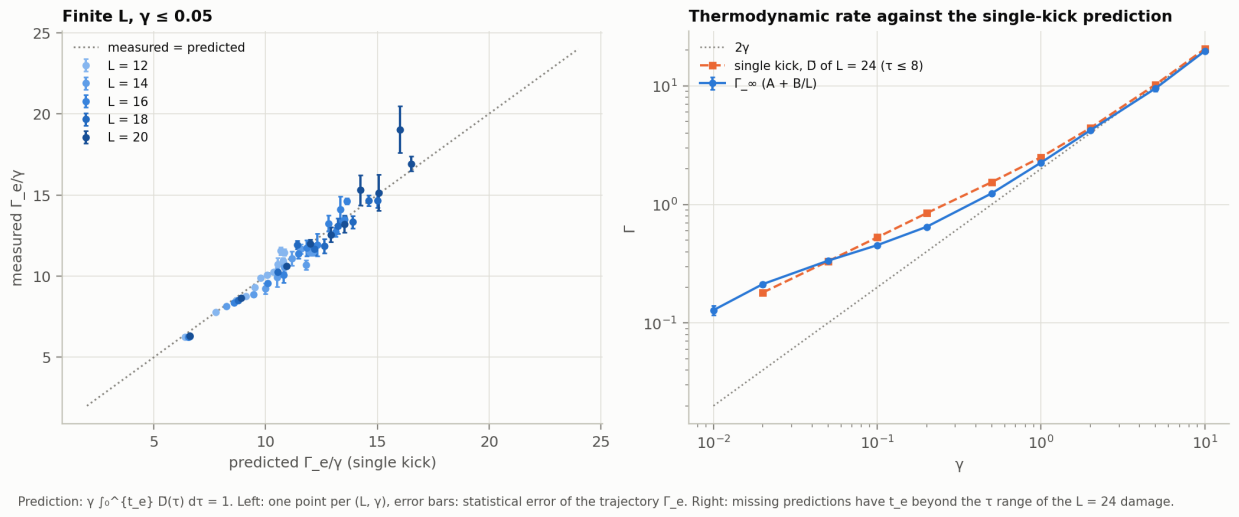


Figure 12: Left: measured against predicted Γ_e/γ at finite L. Right: Γ_∞ against the single-kick prediction from $\bar{D}$ at $L = 24$ (`single_kick.py`).

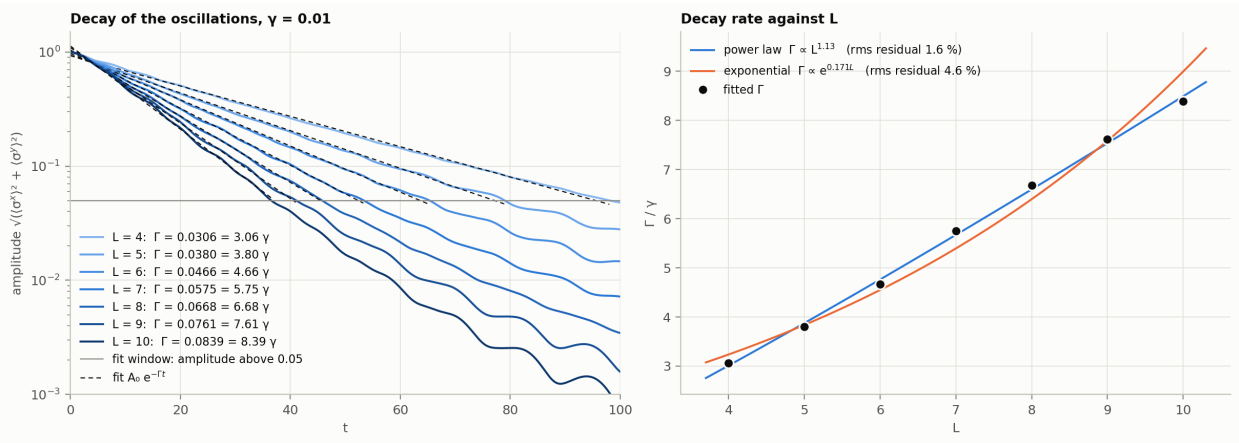


Figure 13: Uniform model, $\gamma = 0.01$: decay of $m_\perp$ from $|-\rangle^{\otimes L}$ for $L = 4, \ldots, 10$ and the fitted rate against L (`plot_scripts/fit_decay.py`).

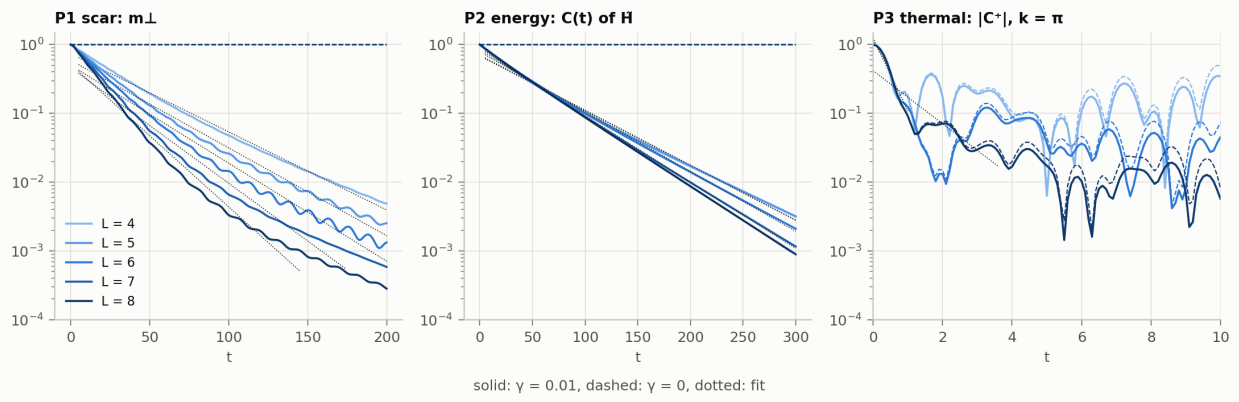


Figure 14: The three probes against time, $\gamma = 0.01$ (solid) and $\gamma = 0$ (dashed); dotted lines are the fits (`plot_scripts/fit_probes.py`).

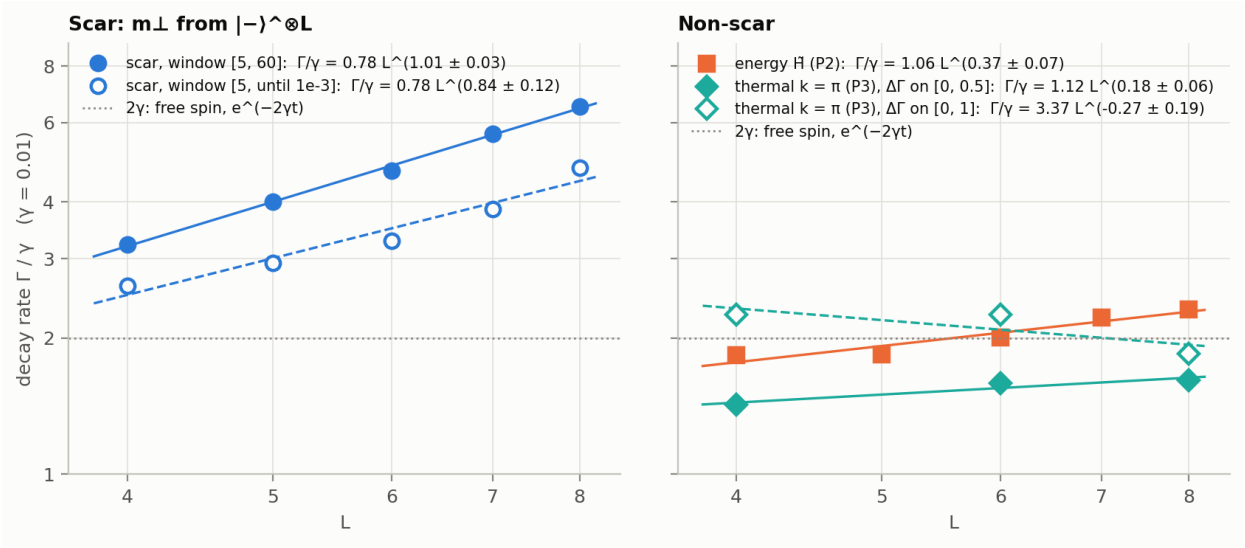


Figure 15: Decay rate against L (log-log) with power-law fits; scar (left) and non-scar probes (right).

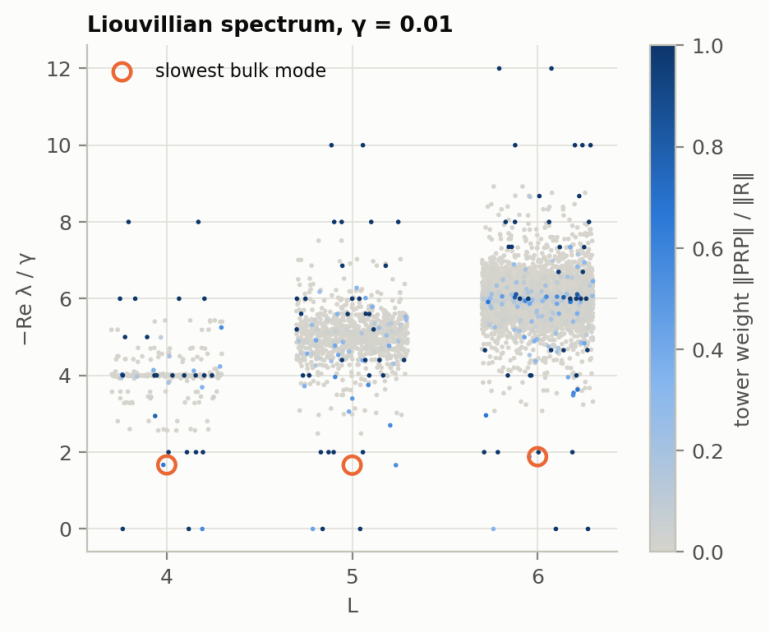


Figure 16: $-\operatorname{Re}\lambda/\gamma$ of all Liouvillian modes, coloured by tower weight; circle: slowest bulk mode (`plot_scripts/plot_spectrum.py`).

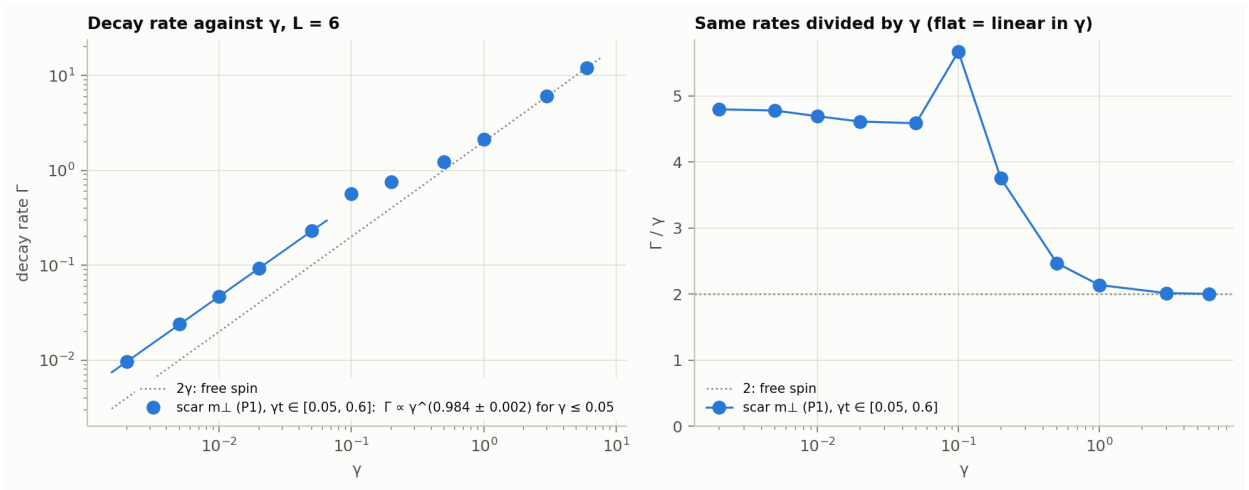


Figure 17: Scar decay rate against γ at $L = 6$; the power law is fitted for $\gamma \leq 0.05$ (`plot_scripts/plot_gamma_scan.py`).

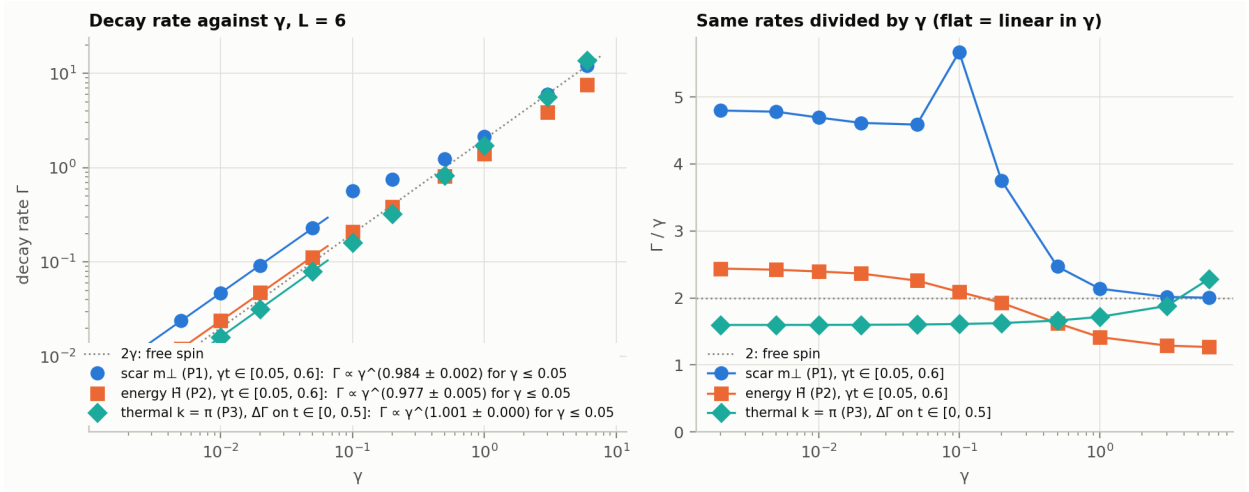


Figure 18: As Fig. 17, with the non-scar probes.